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Incidence of lesions on Fungiidae corals in the eastern Red Sea is related to water temperature and coastal pollution

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ABSTRACT

As sea surface temperatures rise and the global human population increases, large-scale field observations of marine organism health and water quality are increasingly necessary. We investigated the health of corals from the family Fungiidae using visual observations in relation to water quality and microbial biogeochemistry parameters along 1300 km of the Red Sea coast of Saudi Arabia. At large scales, incidence of lesions caused by unidentified etiology showed consistent signs, increasing significantly from the northern to southern coast and positively correlated to annual mean seawater temperatures. Lesion abundance also increased to a maximum of 96% near the populous city of Jeddah. The presence of lesioned corals in the region surrounding Jeddah was strongly correlated with elevated concentrations of ammonium and changes in microbial communities that are linked to decreased water quality. This study suggests that both high seawater temperatures and nutrient pollution may play an indirect role in the formation of lesions on corals.

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1. Introduction

Corals are essential to the biological productivity of reef environments, but are being lost at increasing rates due to factors related to rising global sea surface temperatures (SST) and anthropogenic pressures (Hoegh-Guldberg, 1999; McClanahan et al., 2002; Wilkinson and Souter, 2008). Disease is particularly problematic for corals because it is often non-recoverable (Goreau et al., 1998; Harvell et al., 1999; Willis et al., 2004). Coral disease can be defined as a deviation from normal structure or function, accompanied by a characteristic set of clinical signs (Work and Aeby, 2006). A number of studies have quantified diseased corals, including the presence of lesions or morphologic alterations on the coral, in some cases finding as many as 50% of corals affected on a reef (e.g., Antonius, 1984; Edmunds, 2002; Peters, 1984; Porter et al., 2001; Sato et al., 2009). Understanding the causal factors related to coral disease is complicated, as the holobiont is composed of an assemblage of disparate organisms including the animal host, symbiotic dinoflagellate algae (*Symbiodinium* spp.), bacteria, archaea, fungi and viruses. The degree to which these organisms interact to maintain a healthy, functioning coral is a question of fundamental importance to our understanding of the disease state.

On a global scale, climatic warming is expanding the range and increasing the virulence of certain pathogens (Harvell et al., 2002, 2007; Rosenberg and Ben-Haim, 2002). Anomalously high water temperatures are correlated with outbreaks of coral diseases (Bruno et al., 2007; Harvell et al., 2002; Hayes et al., 2001; Porter et al., 2001; Willis et al., 2004). Understanding the relationship between disease and SST is necessary to predict and mitigate future disease outbreaks in corals and may help pinpoint geographic regions at risk for disease development.

Poor water quality from nutrient pollution or sedimentation is also a major factor linked to declines in coral health (e.g., Acosta, 2001; Bruno et al., 2003; Pastorok and Bilyard, 1985). In particular, moderate increases in nutrient concentrations have been shown to cause a significant increase in severity of coral disease. In experimental field manipulations, the application of fertilizer (>1 μM concentration each of nitrate, ammonium and phosphorus) caused an increase in both the spread of disease and host tissue loss (Bruno et al., 2003). High nutrient levels can also negatively impact coral recruitment, coral cover, and community composition (Fabricius, 2005). In addition to nutrients, abundances and major types of microorganisms are an important component of water quality. For example, ratios of heterotrophic to autotrophic bacteria can indicate an imbalance in an oligotrophic reef environment

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(Dinsdale et al., 2008). Understanding the relationships between coral disease and multiple components of water quality and biogeochemistry is important to quantifying the influence of these factors on coral health.

Finally, aquaculture may be a source of bacterial pathogens to the marine environment and thus may play a role in the distribution of coral disease. Some strains of *Vibrio* spp. have been found to cause disease in aquaculture facilities (Ruangpan and Kitao, 1991) and are highly similar to *Vibrio* spp. found associated with Yellow Band Disease in hard and soft corals (Cervino et al., 2004a, 2008, 2012). Aquaculture facilities are expected to proliferate globally over the next twenty years (Brugère and Ridler, 2004), and their increased abundance may present threats to corals and other marine organisms. The exact relationship between aquaculture effluent and coral health is unknown, and examining the occurrence of diseased corals in relation to known aquaculture facilities may provide evidence for such a link.

The Red Sea provides an ideal environment for understanding the factors influencing coral health because it experiences a wide gradient of temperatures, little terrestrial runoff or freshwater input, and anthropogenic impacts that are almost exclusively isolated, point-source pollution. The temperature gradient (±4 °C mean annual) across the length of the Red Sea allows an opportunity to study reefs residing in substantially warmer versus cooler waters but otherwise with similar properties. In addition, water quality and coral health on individual reefs in close proximity to Jeddah can be studied to assess the impacts of pollution from a highly developed urban coastline. Finally, the waters off the city of Al Lith in the south-central Red Sea region encompass a gradient of distal to proximal reefs subjected to differing amounts of effluent from a large aquaculture facility.

In this study, we characterized the health status of Fungiidae (mushroom corals) in the eastern Red Sea. The Fungiidae family is comprised of scleractinian coral species with free-living and attached life cycles and solitary and colonial morphologies (Hoekesema, 1989). Certain fungiids are capable of active movement across substrate, are competitively aggressive, and are able to regenerate and bud asexually in disturbed habitats (Chadwick, 1988; Gilmour, 2004; Sheppard, 1979). Several studies have found them to be tolerant to sedimentation, despite a life cycle that often ends with the adult coral living on sandy substrate on reef edges (Erftemeijer et al., 2012; Hoekesema, 1989; Schumacher, 1977).

We surveyed reef sites from the straits of Tiran in the north to the Farasan Banks in the south (approx. 1300 km) over the course of three years, to determine how reef environment affects a family of free-living, reef-building coral (Fungiidae: genera as identified in Veron, 2000; *Fungia, Ctenactis, Herpolitha*). These genera were chosen for this study because they were identified during initial surveys as exhibiting consistent patterns of lesions. The lesion patterns were categorized as "Yellow Band Disease-like" due to their visual similarity to lesions described as Yellow Band Disease (YBD) in *Fungia* and *Herpolitha* spp. mushroom corals from Indonesia (Cervino et al., 2008). However, the necessary investigations have not been completed to confirm that this is YBD in the Red Sea. A cultivation-independent microbiological study of *Ctenactis crassa* and *Herpolitha limax* does not suggest a clear pathogen assemblage occurring with the lesioned corals (Apprill et al., 2013), and ongoing work is addressing the histopathological and genetic details of this disease system. The aims of this study were to: (1) quantify the distribution of lesions in fungiid corals throughout the eastern Red Sea; and (2) quantify the relationship between lesion incidence and environmental factors, including sea surface temperature, water quality, and proximity to urbanity and aquaculture facilities.

2. Materials and methods

2.1. Coral surveys

Fungiid populations were surveyed at 56 reefs on the outer shelf of the Red Sea along a 1300 km stretch of Saudi Arabia (Fig. 1). This study was completed over the course of 2008–2010 using five separate cruises: 11–22 November 2008 (central Red Sea), 9–25 June 2009 (southern Red Sea), 1-14 October 2009 (southern Red Sea; with frequent sampling near the aquaculture facility in Al Lith), 18–30 May 2010 (northern Red Sea), and 1–15 September 2010 (northern Red Sea) (Supplementary Table 1). Three coral genera from the family Fungiidae were included in the investigation: *Fungia, Herpolitha,* and *Ctenactis* (taxonomy according to Veron, 2000). Belt transect surveys of 50 m × 1 m (Hoeksema, 2012a,b) were completed at 10 m, the average depth of reefs in the study area using SCUBA, and 2–4 transects were assessed per reef. All fungiids within transects were identified to genus and categorized as healthy, bleached or lesioned.

Healthy fungiids showed no visible signs of stress (Fig. 2A, B). The coral tissue was typically a deep brown color that paled towards the edges of the septal ridges. Bleached fungiids frequently exhibited a white mottled pattern (Fig. 2C) or were uniformly pale (Fig. 2D). When mottled, the bleaching areas appeared haphazardly located on the surface of the coral, with indistinct edges between

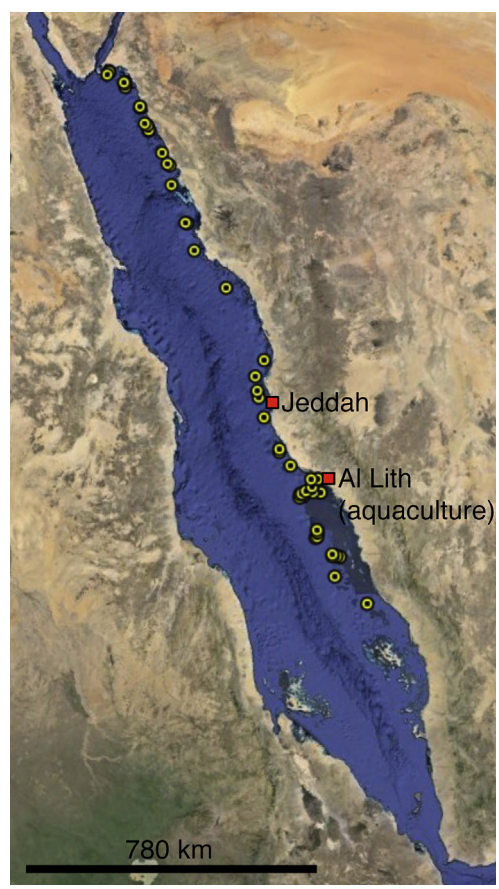

Fig. 1. Map of the reef sites spanning ~1300 km that were surveyed for coral health in the Red Sea. The large urban area of Jeddah, and the large aquaculture facility at Al Lith, are indicated. Image is courtesy of Google Earth (2010) under the academic print distribution guidelines, and include data from SIO, NOAA, U.S. Navy, NGA and GEBCO and images from U.S. Geological Survey, Cnes/Spot Image and 2001 DigitalGlobe

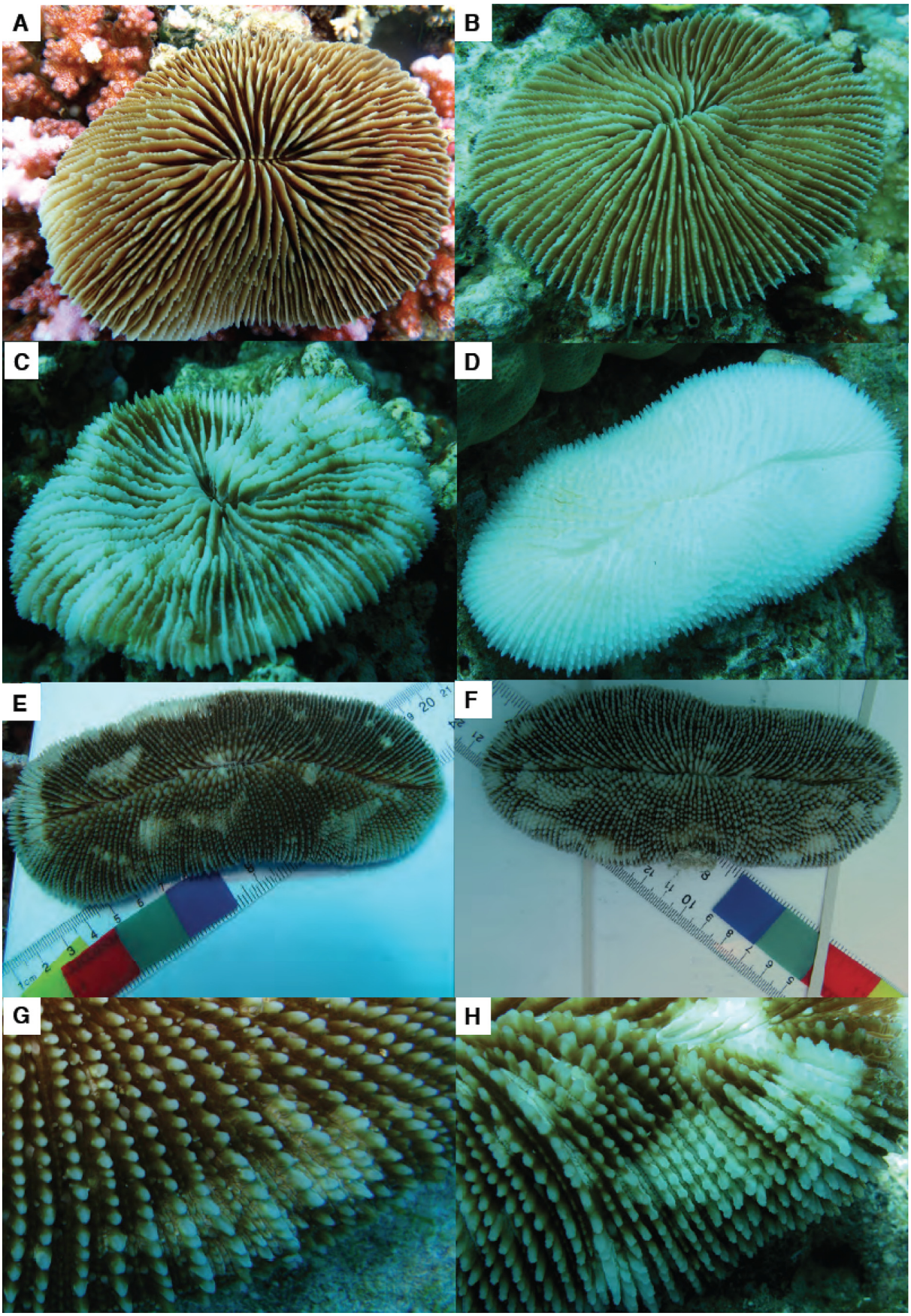

Fig. 2. Photographs of fungiid corals of various health status, including healthy (A, B), bleached (C, D) and lesioned (E, F). Close up photographs of typical lesions on fungiid corals (G, H).

healthy and bleached areas. Fungiids with a yellow blotch pattern similar to those found by Cervino et al. (2008) were classified as 'lesioned' (Fig. 2E, F). Macro photos illustrate the margins of the lesions and extent of tissue damage (Fig. 2G, H). Characteristics of fungiid lesions included: central and peripheral location, multifocal to coalescing distribution, lanceolate to irregular shape with distinct edges and smooth margins, and a pale yellow discoloration. The lesions are two-dimensional and small in size relative to coral (or polyp) size. The number of lesions per coral varied. Symptoms of tissue loss without algal growth, as well as skeletal degradation, were observed occasionally but not quantified in this study.

2.2. Water sampling

Seawater was sampled for measurements of inorganic nutrients and microbial abundances at sites during the 2009 and 2010 cruises (Supplementary Table 1). Seawater was sampled just above the depth of corals, approximately 10 m, using a Masterflex peristaltic pump (Cole Parmer, Vernon Hills, IL, USA). For inorganic nutrients, water was collected from the same depth into 150 ml polypropylene acid-washed bottles and frozen at −20 °C. Samples (1 ml) for microbial abundances were fixed in a final concentration of 1% (v:v) paraformaldehyde and stored in cryovials in liquid nitrogen for 3 weeks, followed by storage at −80 °C until analysis.

2.3. Nutrient analysis

Dissolved inorganic nutrient concentrations (ammonium, nitrate + nitrite, phosphate and silicate) were measured using a continuous segmented flow system consisting of a Technicon AutoAnalyzer II (SEAL Analytical, Mequon, WI, USA) and an Alpkem RFA 300 Rapid Flow Analyzer (Alpkem, Clackamas, OR, USA). Phosphate was measured using a modified molybdenum blue method (Bernhardt and Wilhelms, 1967). Standard methods were utilized to measure nitrate + nitrite (Armstrong et al., 1967). Ammonium was measured using the indophenol blue method (U.S. Environmental Protection Agency, 1983), and concentrations were verified using the method of Holmes et al., 1999.

2.4. Direct cell counts

Microbial abundances were determined using flow cytometry. In order to enumerate both pigmented and non-pigmented cells, aliquots of the preserved water samples were analyzed in two manners, stained and unstained. Unstained samples were run on an EPICS ALTRA flow cytometer (Beckman Coulter Inc., Brea, CA, USA), and excitation in the visible wavelengths was used to enumerate cyanobacteria (*Prochlorococcus* and *Synechococcus* spp.) and eukaryotic phytoplankton (picoplankton), on the basis of chlorophyll (red fluorescence, 680 nm), phycoerythrin (orange fluorescence, 575 nm), forward scatter, and 90° side scatter signatures. A second aliquot of sample was diluted 1:10 into 30 mM (final) potassium citrate buffer, and stained with Sybr Green I (1:5000 final dilution of initial stock) (Molecular Probes, Eugene, OR, USA) for two hours in the dark at 4 °C. Excitation at 488 nm on the same machine was used to enumerate picoplankton on the basis of DNA staining (Sybr Green I green fluorescence, 525 nm), chlorophyll (red fluorescence), phycoerythrin (orange fluorescence), forward scatter, and 90° side scatter signatures, and counts of *Prochlorococcus* spp. cells from unstained samples were subtracted from total prokaryotic cells (as indicated by their DNA signature) to obtain abundances of non-pigmented picoplankton. Data were analyzed off-line using FlowJo software (v. 6.3.3, Tree Star, Inc., Ashland, OR, USA).

2.5. Sea surface temperature

Sea surface temperature (SST) data were obtained from the MODIS (MODerate Resolution Imaging Spectroradiometer) sensors onboard the NASA Aqua platform using mid-infrared (IR) and thermal IR channels (Brown and Minnett, 1999; Walton et al., 1998). SST data used in this study were produced with the Giovanni online data system, developed and maintained by the NASA GES DISC (Acker and Leptoukh, 2007). The SST data were acquired at 9 km spatial resolution, with averages computed over three years, corresponding to the same period as the coral health surveys (October 2007–2010).

2.6. Statistical analysis

A principal component analysis (PCA) was used to explore relationships between environmental parameters directly measured at each site, as well as latitude and SST. The analysis did not include information about the lesioned corals, and these data were overlaid onto the symbols for each site. The analysis was conducted using PRIMER version 6.1.13.

Linear regressions were performed to quantify the relationships between coral lesion incidence and environmental variables including SST, inorganic nutrient concentrations, microbial communities, and distance from point sources including a major urban area (Jeddah) and aquaculture facility (Al Lith). Relationships that did not follow normal distributions were further explored using data transformations, and square root and logarithmic transformations of data were found to minimize non-normal residuals and therefore the transformed data were presented in the relevant figures. Analyses were conducted using StatPlus version 5.8.0, 2009.

3. Results

3.1. Fungiid health in the eastern Red Sea

Individual fungiid corals were characterized as healthy, bleached or lesioned along a 1300 km transect in the eastern Red Sea, and the proportions of corals displaying these health categories varied greatly between sites (Fig. 3). Bleaching of free-living corals could be related to multiple factors including irritation by sediment and inversion on the seafloor, as well as abnormal temperatures (Schumacher, 1977). For this reason, although the study quantified bleaching, bleaching was not investigated further. Lesioned corals were further investigated, and lesions were found to affect 27% of fungiids throughout the reefs surveyed. On the

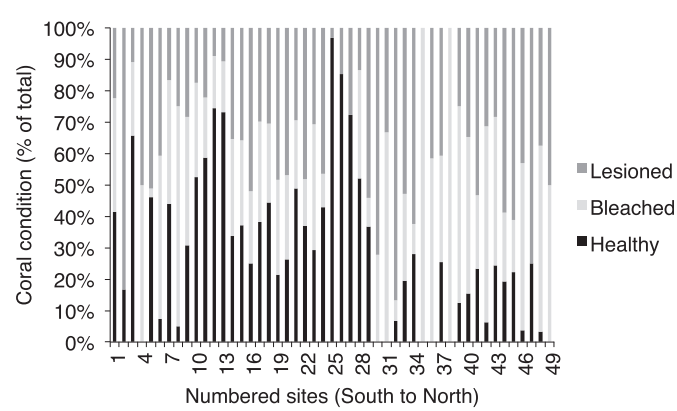

Fig. 3. Percentages of fungiid corals categorized as healthy, bleached or lesioned at survey sites throughout the eastern Red Sea (2–4 transects per site).

broadest spatial scale, lesioned fungiids were more prominent on southern compared to northern Red Sea reefs (Fig. 4), and this trend is statistically significant (Mood median test, Chi-squared = 214 15.96, DF = 3, $p = 0.001$). On smaller scales, particularly in the central Red Sea, lesioned fungiids peaked to incidences of 97%, and were more prevalent north of the city of Jeddah, a major Saudi Arabian city on the coast (Fig. 4).

3.2. General properties of nutrients, microbial abundances and SST in the eastern Red Sea

Seawater inorganic nutrient concentrations were measured at 45 of the 52 sites surveyed for coral health (Supplementary Table 1). Concentrations of phosphate ranged from 0.021 to 0.132 μM, (mean 0.062 μM), and were generally lower at the northernmost sites (Supplementary Fig. 1A). Concentrations of nitrate + nitrite ranged between 0.004 and 0.528 μM (mean 0.181 μM) and were generally < 0.2 μM at the northernmost sites (Supplementary Fig. 1B). Ammonium concentrations were mostly below detection, ranging from 0.004 to 2.972 μM, with a mean concentration of 0.169 μM (Supplementary Fig. 1C). Elevated concentrations of ammonium, 0.26–1.3 μM, were detected at three sites just north and down current of the city of Jeddah. Silicate at all sites ranged from 0.305 to 1.152 μM with a mean concentration of 0.793 μM (Supplementary Fig. 1D).

Abundances of the major microbial groups were also measured at 38 of the 52 survey sites as an indirect indicator of water quality (Supplementary Table 1, Supplementary Fig. 2). In the water above the reefs surveyed, heterotrophic bacteria ranged from 3.6×10^5–1.4×10^6 cells ml^{-1} (mean 5.8×10^5 cells ml^{-1}), with the highest concentrations at the southernmost site of Sumayr (Supplementary Fig. 2A). Concentrations of *Synechococcus* varied from 2.6×10^3–2.2×10^5 cells ml^{-1} (mean 5.0×10^4 cells ml^{-1}), with a trend of fewer cells at the northern latitude reefs (Supplementary Fig. 2B). *Prochlorococcus* cells ranged from undetectable to 1.1×10^5 cells ml^{-1} (mean 4.6×10^4 cells ml^{-1}) throughout the surveyed reefs (Supplementary Fig. 2C). Reefs where *Prochlorococcus* were undetectable were located in the central Red Sea, offshore and directly north of the city of Jeddah (spanning 21.6–22.4°N latitude). The remaining sites contained at least 21,000 *Prochlorococcus* cells ml^{-1}. Picoeukaryote abundance ranged from $90 - 7.5 \times 10^3$ cells ml^{-1} (mean 3.1×10^3) at all sites, with similar concentrations throughout the basin (Supplementary Fig. 2D).

Prokaryotic metabolism, indirectly assessed using the ratio of heterotrophic to autotrophic (*Prochlorococcus* + *Synechococcus*) bacterial cells, exhibited a large range at the surveyed sites, 1.2–27.8 (mean 8.0). Ratios were generally lower (increased autotrophy) in the southern Red Sea (Supplementary Fig. 2E). Ratios at one site in the southern Red Sea (AQ3 at 19.1°N) was especially low, < 1.5, and one site in the northern Red Sea (Pisces I, 27.3°N) was exceptionally high, with a cellular ratio of 27.8.

SST was not directly measured at each site, but satellite data revealed that mean-annual SST averaged over three years for each study site ranged from 26 °C to 29.5 °C (Supplementary Fig. 3). The highest temperatures were concentrated in the southern region.

3.3. Relation of lesioned corals to Red Sea environmental conditions

A principal component analysis (PCA) examined the relationship between the different environmental parameters measured at each site, as well as latitude and SST. This analysis indicated that sites with high numbers of lesioned corals (>60%) were associated with the PC2 axis, and were related to increasing concentrations of ammonium, as well as decreasing abundances of silicate, *Prochlorococcus* spp. and heterotrophic bacteria (Fig. 5). Sites with minimal percentages of lesioned corals associated more with the PC1 axis, and were related to higher latitude, lower SST, and decreased abundances of *Synechococcus* spp. Thus, SST, microorganisms, inorganic nutrients and geographic latitude were related to incidences of fungiid lesions across the eastern Red Sea.

The environmental parameters that were particularly identified as variables exhibiting a relationship with the presence or absence of lesioned corals were explored individually in more detail. To examine SST as a potential factor influencing fungiid lesions, three-year mean SST was examined against percentage of lesioned corals at each site. A significant relationship exists, with higher incidences of lesions generally occurring in the warmer water ($r^2 = 0.25555$, $p = 0.0021$ $n = 52$; Fig. 6).

Regression and correlation analysis of relationships between lesioned corals and microbial cellular abundance revealed that significant interactions exist for *Prochlorococcus* spp. ($r^2 = 0.20527$, $p = 0.0049$, $n = 38$; Fig. 7A), *Synechococcus* spp. ($r^2 = 0.21463$, $p = 0.004$, $n = 38$; Fig. 7B) and picoeukaryote abundances ($r^2 = 0.17239$, $p = 0.0106$, $n = 38$; Fig. 7C), compared to occurrence of coral lesions. The regression analysis also indicated that

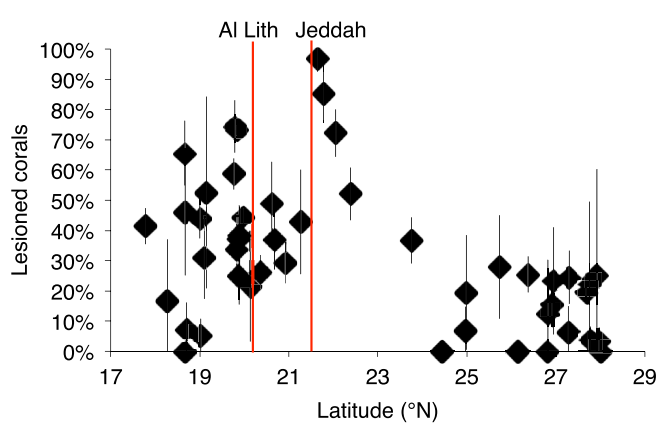

Fig. 4. Distribution of lesioned corals at the survey sites (2–4 transects per site with standard error bars). The locations of Jeddah (large urban area) and Al Lith (large aquaculture facility) relative to the sampling sites are indicated.

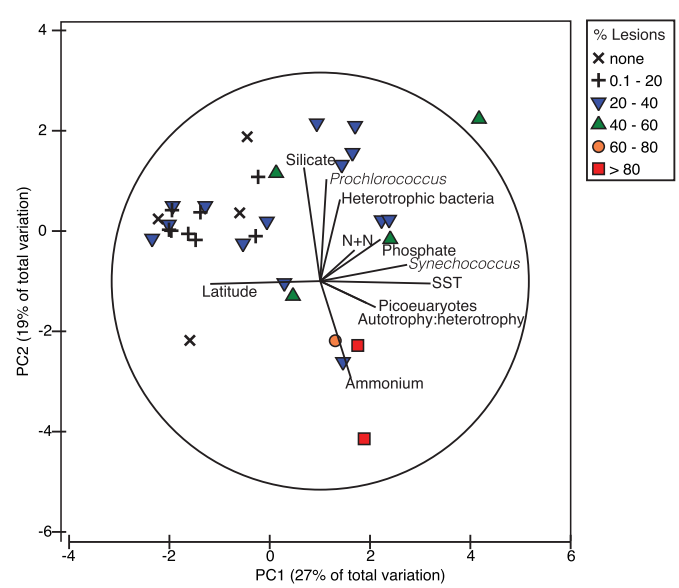

Fig. 5. Principal component analysis (PCA) of the measured environmental variables at each sampling site, in relation to the percentage of lesioned corals. The vector overlays represent multiple correlations between ordination axes and environmental variables. Only the sites with complete environmental data sets are represented ($n = 32$).

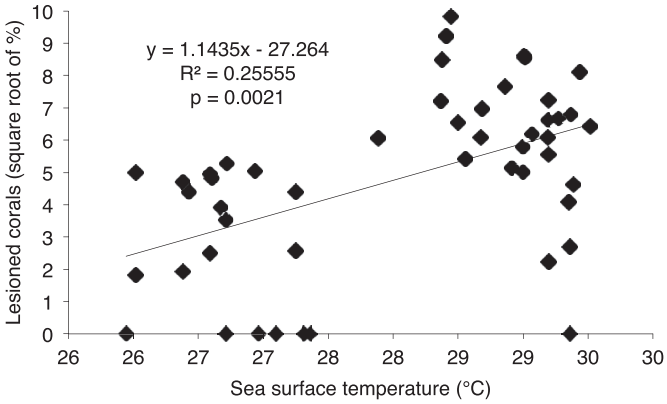

Fig. 6. Comparison of percent abundance of lesioned corals (square root transformed) compared to three-year mean SST ($n = 52$).

Prochlorococcus spp. decreased as lesion occurrences increased, whereas *Synechococcus* spp. and picoeukaryote abundances increased with increasing percentage of lesions. There were no relationships between abundances of heterotrophic bacteria and lesioned corals (Fig. 7D), and the ratio of heterotrophic to autotrophic prokaryotes was also not significantly related to lesioned corals (Fig. 7E).

Similar to the findings for the PCA analysis, regression analysis of lesioned corals compared to concentrations of ammonium ($r = 0.21835$, $p = 0.004$, $n = 38$; Fig. 8A) and silicate ($r = 0.13991$, $p = 0.025$, $n = 38$; Fig. 8B) were found to be significant. Higher ammonium concentrations were associated with greater percentages of lesions, and the opposite relationship was found for silicate (low silicate when lesions were more prevalent). However, it should be noted that although the relationship between ammonium and lesioned corals was significant, at many sites concentrations were below detection and this trend was only apparent at the local scale near the Red Sea's largest city of Jeddah. No significant relationship was found between the percentages of coral lesions at each site and concentrations of phosphate (Fig. 8C) or nitrate and nitrite (Fig. 8D).

3.4. Urban and aquaculture influences on coral lesions

The nutrient and microbial data in the eastern Red Sea suggest that there may be specific point-source impacts from the surrounding terrestrial environment. Lesioned corals were prominent south of Al-Lith, a city supporting a low population (<130,000 people; Central Department of Statistic) but a large shrimp aquaculture facility. The potential impact of aquaculture pollutants on lesioned corals was assessed using regression analysis with distance from the aquaculture outfall, and did not demonstrate a significant relationship (Fig. 9A).

A relationship between distances from the major urban center of Jeddah (population 3.4 M; Central Department of Statistics) was found using regression analysis. Specifically, abundances of lesioned corals were significantly related to distance from Jeddah (Fig. 9B). Populations of the other eastern Red Sea cities were considerably lower than Jeddah (25,568–233,236 people; Central Department of Statistics) and were not further investigated.

3.5. Urban water quality related to lesioned corals near Jeddah: a summary

The relationships between lesioned corals and water quality and proximity to Jeddah were further examined on a scale relative to point source pollutants. Oceanographic currents along the eastern Red Sea shore generally move from the south to the north, and both concentrations of ammonium and lesion occurrence reached maximum values at the first site north of Jeddah

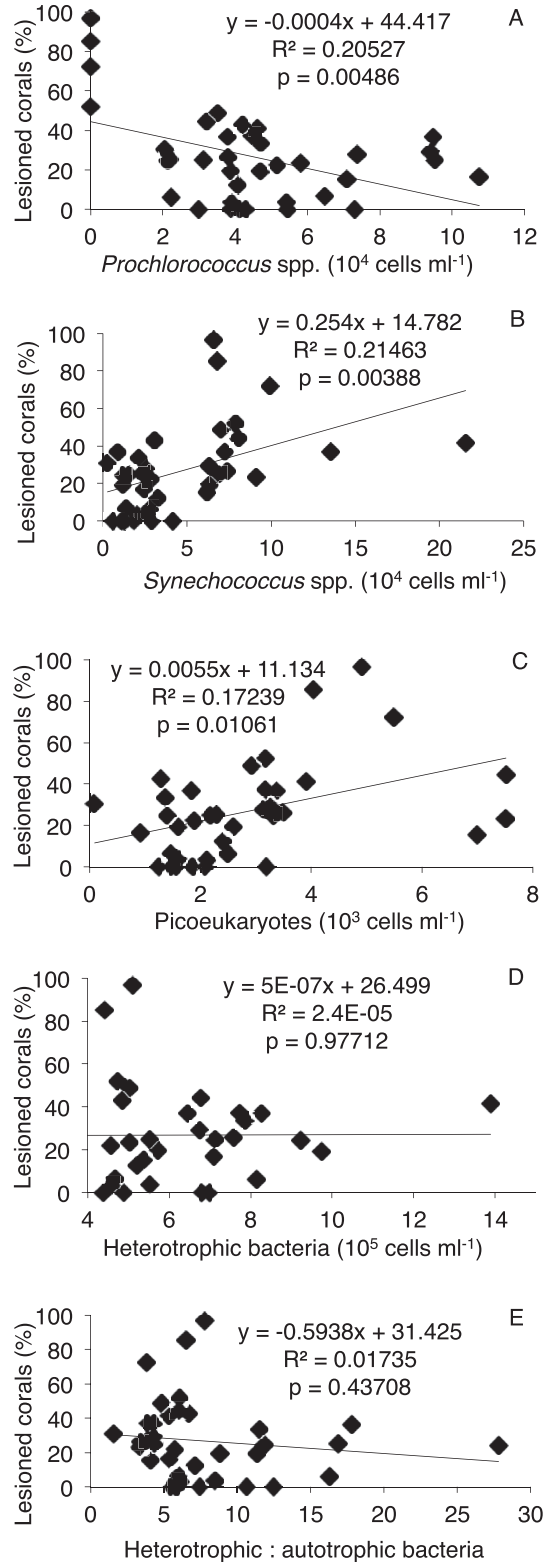

Fig. 7. Regressions of microbial cellular abundances against percentage of lesioned corals, at sites surveyed for coral health, including *Prochlorococcus* spp. (A), *Synechococcus* spp. (B), picoeukaryotes (C), heterotrophic bacteria (D), as well as the ratio of heterotrophic to autotrophic bacteria regressed against lesioned corals (E) ($n = 38$).

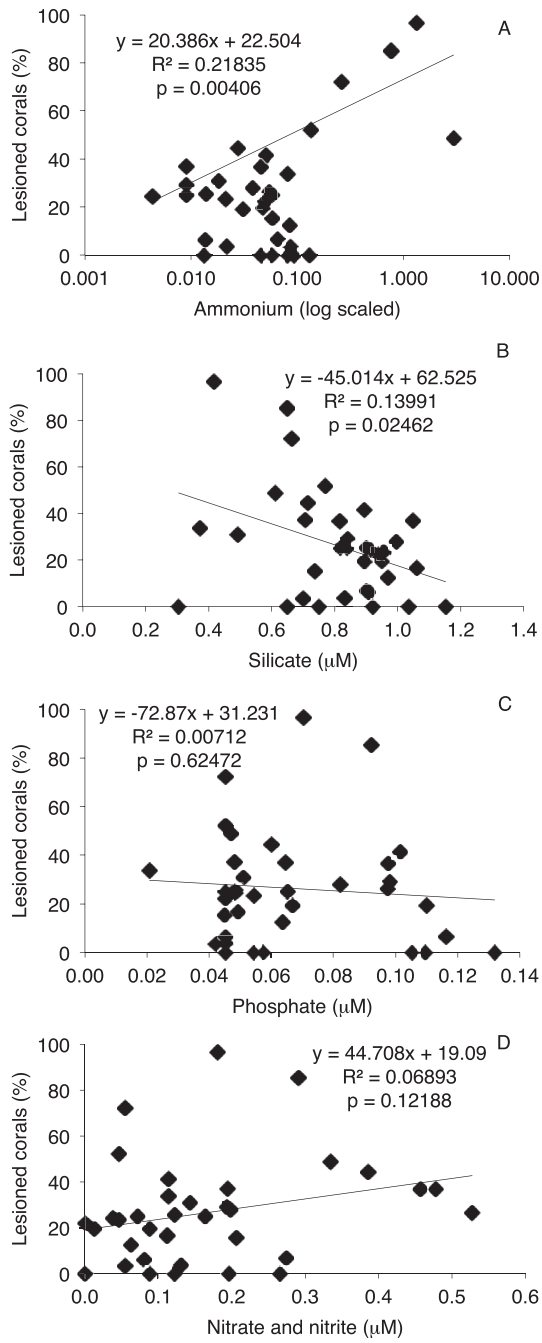

Fig. 8. Regressions of the major inorganic nutrients ammonium (A), silicate (B), phosphate (C) and nitrate + nitrite (D) against percentage of lesioned corals ($n = 38$).

(~40 km), and then decreased with distance northward (Fig. 10). In the same sites immediately north of Jeddah, the oligotrophic bacteria *Prochlorococcus* spp. were absent. The lack of *Prochlorococcus* spp. cells near Jeddah coincides with the comparatively high concentrations of coastal-derived ammonium, which are orders of magnitude lower at all other reef sites examined. Collectively, these data provide evidence for decreased water quality (elevated ammonium, undetectable oligotrophic bacteria (*Prochlorococcus* spp.) and increased occurrences of coral lesions near the city of Jeddah; each signal diminishes with distance north from Jeddah.

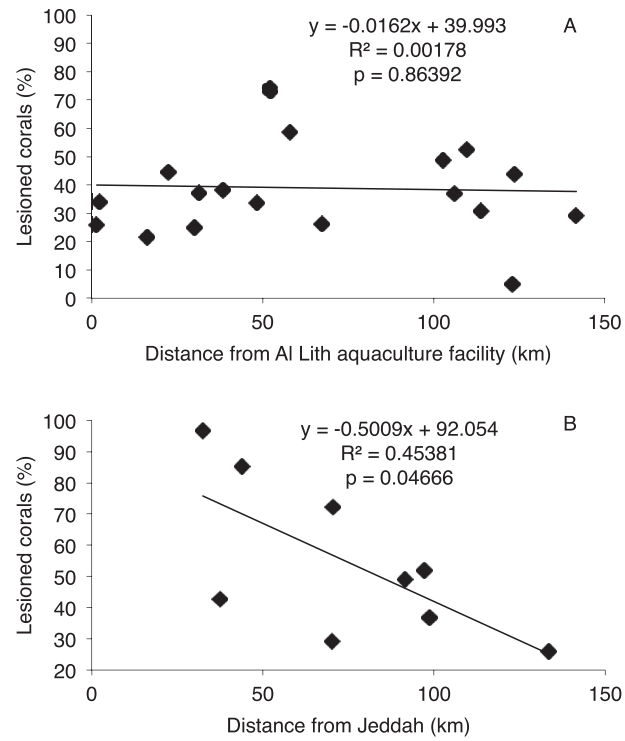

Fig. 9. Regression analysis comparing the percentage of lesioned corals to distance from the aquaculture facility in Al Lith (A) and the city of Jeddah (B).

4. Discussion

4.1. Prevalence of lesioned fungiids

The prevalence of lesioned *Fungiidae* corals along the eastern coast of the Red Sea was high (average 27%, peak prevalence of 97%) compared to previous quantifications of coral lesions in the Caribbean and Great Barrier Reef (Green and Bruckner, 2000; Porter et al., 2001; Sato et al., 2009; Willis et al., 2004). These previous studies report 5–65% incidence of colonies affected by black band disease (BBD), white diseases, and YBD. In this Red Sea study, higher prevalence of lesioned corals was correlated to factors including seawater temperature and water quality, reflected by inorganic nutrients and microbial abundances. These data suggest

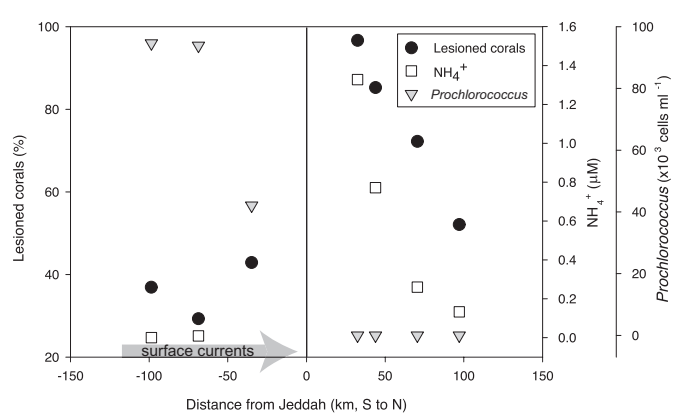

Fig. 10. Summary diagram displaying the relationship between the percentage of lesioned fungiids at sites north and south of Jeddah, and concentrations of ammonia and *Prochlorococcus* spp. cells in the water at each site. The direction of coastal currents is also represented, generally traveling from S to N.

that both regional climate (temperatures) and local pressures from urban areas affect the health of Red Sea corals.

4.2. Sea surface temperature (SST) related to lesioned corals

There was a clear pattern of higher prevalence of lesions on the mushroom corals in the southern Red Sea. The southern Red Sea typically experiences 1.5–2.5 °C higher mean annual SST than the northern Red Sea. In particular, higher incidences of lesions were found in southern and central Red Sea reefs, and bleaching was a common phenomenon affecting fungiids in the northern reefs.

The relationship between elevated SST and increased lesions and/or coral disease is well documented (Alker et al., 2001; Cervino et al., 2004a,b, Cervino et al., 2008; Kushmaro et al., 1998; Selig et al., 2006; Toren et al., 1998). Studies by Ben-Haim and et al. (2003a,b) have suggested that this relationship occurs because bacterial pathogens involved in the disease state are more virulent with high temperatures. Elevated temperatures could also impair coral immunity, rendering them more susceptible to infection (Porter et al., 2001). The continued warming expected for the future may have detrimental impacts on corals, including fungiids in the Red Sea. The sources of the pathogens as well as temperature-related impacts on coral disease are critical areas for future research.

Additional longer-term, large-scale studies are needed to monitor corals over time and differentiate between patterns of bleaching and lesions. It should be noted that at depths greater than 6 m, fungiid corals will sometimes exhibit mottled patterns of bleaching in response to thermal stress (Hoeksema, 1991). However, the pattern of bleaching described by Hoeksema (1991) differs from the lesion pattern observed in the Red Sea. Tolerance to bleaching can vary across depth and by species (Hoeksema, 1991; Hoeksema and Matthews, 2011).

Shorter-term surveys of coral health can often overlook unseen variables (Edmunds and Bruno, 1996; Porter et al., 2001). For example, continued monitoring of the lesioned states would aid in determining if the condition is chronic. Seasonality and water temperature effects on the coral lesions could also be examined to confirm the sensitivity of lesions to SST. By continuing comparative regional studies of coral reef communities, particularly in the unique yet relatively poorly studied Red Sea, we may gain valuable insights into future coral health impacts from rising sea surface temperature as well as from local anthropogenic influences (Berumen et al., 2013).

4.3. Urban water quality related to lesioned corals near Jeddah

Several measurements of water quality proximal to the city of Jeddah were related to the presence of lesioned corals. Specifically, high incidences of lesioned corals were reported at sites corresponding to elevated concentrations of ammonium, often a signature for sewage or manure runoff in coastal areas (Duedall et al., 1975, Risk et al., 2009). In the same sites immediately north (down current) of Jeddah, *Prochlorococcus* spp. were absent. While *Prochlorococcus* spp. are able to utilize ammonium (and not nitrate) as a nitrogen source (Garcia-Fernandez et al., 2004), populations generally decrease or are absent in more eutrophic and brackish regions (Partensky et al., 1999). Studies of an urban coastal environment in Kaneohe Bay, Hawaii, report undetectable concentrations of *Prochlorococcous* spp., but detectable *Synechocccus* spp., inside the Bay (Apprill and Rappé, 2011; Cox et al., 2006). The lack of *Prochlorococcus* spp. cells near Jeddah may be related to the comparatively higher concentrations of coastal-derived ammonium, which are orders of magnitude lower at other reef sites, and/or other coastal pollutants, such as toxins and pharmaceutical products, that were not measured. The signatures for low water quality are detected north and not south of Jeddah and are most likely explained by the general direction of surface current movement when these measurements were taken (Quadfasel, 2001), as well as the fact that urban populations are focused in northern Jeddah (Vincent, 2003).

Jeddah's population is nearing three and a half million people, and has experienced high growth over the last decade (Central Department of Statistics). A number of environmental issues contribute to coastal runoff, including direct sewage discharge and the high level of cesspool-influenced groundwater leaking into coastal lagoons (Vincent, 2003). The results presented here agree with previous studies demonstrating declines in coral health near human population centers (e.g., Bruno et al., 2003; Green and Bruckner, 2000; Harvell et al., 1999; Pastorok and Bilyard, 1985; Porter et al., 2001). In previous studies of the Gulf of Aqaba, Red Sea and Kaneohe Bay, Hawaii, survivorship of corals was directly correlated with distance from sewage outfall (Smith et al., 1981; Walker and Ormond, 1982).

Coral disease has been linked with high nutrient levels, including but not limited to ammonium. Bruno et al. (2003) found that adding a minimum of 1.0 μM nitrate, 0.9 μM phosphorus, and 1.0 μM ammonium could significantly increase incidence of some marine diseases. The moderate change in ammonium observed by Bruno et al. (2003) was approximately one third of the maximum concentrations observed on reefs in the eastern Red Sea. Therefore, the elevated ammonium and subsequent alteration in the microbial community may be reflecting or contributing to the increasing stress on fungiids occurring near Jeddah.

Free-living scleractinians are unique in that their life history may include a phase in which they live on sandy substrate. Fungiids that are free-living often occur near the reef base and will survive limited flipping or burial in sediment (Chadwick, 1988; Chadwick-Furman and Loya, 1992; Goffredo and Chadwick-Furman, 2000). Further study is needed on the sediment chemistry, as well as suspended sediment levels of the Red Sea, as this may be a factor affecting the health of these corals. However, fungiids have been found to be resilient to bleaching and sedimentation (Bongaerts et al., 2012; Furby et al., 2013; Schumacher, 1977). The *Fungia, Herpolitha* and *Ctenactis* corals studied here are seemingly unique in their vulnerability to the sources of stress investigated in this study, and they may represent new indicator genera for coral stress within the Red Sea.

4.4. Aquaculture outfall not related to lesioned corals

One of the largest shrimp aquaculture facilities in the world is near the coast in Al Lith. This facility's effluent canal is less than a kilometer from several sites surveyed in this study. Despite the high sampling of reefs near Al Lith, nutrient levels at sites near the facility were not anomalously high, nor was the frequency of lesioned corals. Over 2 km away from fish pen aquaculture in the Philippines, total inorganic nitrogen concentrations are over 14 μM (Garren et al., 2008). This is approximately an order of magnitude higher than concentrations measured <1 km from the Al Lith facility outfall. Overall these results and the low incidences of lesioned corals surrounding Al Lith suggest that the aquaculture facility outfall may not be a major source of stress to fungiid corals in this region. A comprehensive survey of the abundance, distribution and health of diverse coral species as well as water quality parameters would be necessary to address the impact of the Al Lith aquaculture outfall on corals.

4.5. Nutrients and microbial abundances in the eastern Red Sea

The inorganic nutrient and microbial measurements presented here are the most comprehensive dataset available on water

quality for the eastern Red Sea, a relatively understudied region (Berumen et al., 2013). Reef water microbial and biogeochemical loops have important implications for the health of corals (Dinsdale et al., 2008; Garren et al., 2008; Pastorok and Bilyard, 1985). However, there are a number of caveats associated with this dataset, most notably the fact that water samples were collected during different years, seasons and tidal fluctuations, due to availability of ship time. However, these data provide some important and novel observations about Red Sea microbial biogeochemistry.

Throughout the Red Sea, concentrations of inorganic nutrients were quite low; average total dissolved inorganic nitrogen (DIN) was generally less than 0.55 μM. Levels of ammonium rarely exceeded detectable ranges at nearly all but the Jeddah-proximal sites. These concentrations are lower than detected for other coral reef environments, where DIN ranges from 1 to 4 μM or more (Apprill and Rappé, 2011; Dinsdale et al., 2008; Szmant, 2002). Thus, the observed ammonium enrichment near Jeddah may be specifically problematic for Red Sea corals compared to corals regularly exposed to higher concentrations.

Microbial metabolism exhibited a latitudinal relationship in the eastern Red Sea, and the increased autotrophy in the southern Red Sea is likely related to elevated seawater temperatures in this region. Both *Prochlorococcus* and *Synechocccus* spp. generally exhibit higher growth rates in warmer waters (Moore et al., 1995), although the ecotypes of these species have not been well studied in the Red Sea. Abundances of heterotrophic bacteria at the study sites were similar to previous measurements in the Red Sea (Weisse, 1989). However, concentrations of heterotrophic bacteria, *Synechococcus* spp. and picoeukaryotes were 0.5–1 order of magnitude lower than coastal Hawaiian reefs (Apprill and Rappé, 2011). Interestingly, although nutrient levels are higher in the Line Island Reefs in the central Pacific (Dinsdale et al., 2008), microbial cell abundances (heterotrophic and autotrophic bacteria) were comparatively higher in the Red Sea. These limited results suggest that the microbial biogeochemistry of the Red Sea may be different than other open-ocean reef environments, possibly due to the limited circulation and water exchange and the oligotrophic nature of the Red Sea region.

5. Conclusions

This study represents a novel assessment of eastern Red Sea mushroom corals along 1300 km of Saudi Arabian coastline, where coral health, nutrient enrichment, and seawater microbial communities were previously uncharacterized on a comparable scale. The correlation of coral lesions with SST was the most significant relationship observed throughout the eastern Red Sea. Elevated sea surface temperatures may serve to increase the virulence of pathogens or diminish corals' immune defenses. The relationship between ammonium and the prevalence of fungiid corals with lesions was significant at the local scale near the Red Sea's largest city, Jeddah. Ammonium is an indicator of sewage pollution, and together with the lack of *Prochlorococcus* spp., reflects poor water quality in the region immediately down current from Jeddah. The relationship between coral health and proximity to aquaculture effluent was also examined, but no correlations were found. More focused studies would be necessary to draw conclusions regarding the potential impact of pathogens from aquaculture outfall and coral health. Overall, this large-scale geographic study provides an opportunity to examine the relationship between environmental factors and coral health. As climate change continues to impact marine ecosystems, studies of Red Sea corals will be critical for understanding how coral reefs function under high temperature and increasing anthropogenic impact.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.marenvres.2014.04.002.

References

Acker, J.G., Leptoukh, G., 2007. Online analysis enhances use of NASA earth science data. Eos, Trans. Am. Geophys. Union 88, 17.
Acosta, A., 2001. Disease in zoanthids: dynamics in space and time. Hydrobiologia 460, 113–130.
Alker, A.P., Smith, G.W., Kim, K., 2001. Characterization of *Aspergillus sydowii* (Thom et Church), a fungal pathogen of Caribbean sea fan corals. Hydrobiologia 460, 105–111.
Antonius, A., 1984. Human impact on corals in fringing reefs near Jeddah, Red Sea. In: Proceedings Symposium Coral Reef. Environ. Red. Sea Jeddah, pp. 363–389.
Apprill, A., Rappé, M.S., 2011. Response of the microbial community to coral spawning in lagoon and reef flat environments of Hawaii, USA. Aquat. Microb. Ecol. 62, 251–266.
Apprill, A., Hughen, K., Mincer, T., 2013. Major similarities in the bacterial communities associated with lesioned and healthy Fungiidae corals. Environ. Microbiol. 15, 2063–2072.
Armstrong, F.A.J., Stearns, C.R., Strickland, J.D.H., 1967. The measurement of upwelling and subsequent biological processes by means of the Technicon Auto-Analyzer and associated equipment. Deep-Sea Res. 14, 381–389.
Ben-Haim, Y., Thompson, F.L., Thompson, C.C., Cnockaert, M.C., Hoste, B., Swings, J., Rosenberg, E., 2003a. *Vibrio coralliilyticus* sp. nov., a temperature- dependent pathogen of the coral *Pocillopora damicornis*. Int. J. Syst. Evol. Microbiol. 53, 309–315.
Ben-Haim, Y., Zicherman-Keren, M., Rosenberg, E., 2003b. Temperature-regulated bleaching and lysis of the coral *Pocillopora damicornis* by the novel pathogen *Vibrio coralliilyticus*. Appl. Environ. Microbiol. 69, 4236–4242.
Bernhardt, H., Wilhelms, A., 1967. The continuous determination of low-level iron, soluble phosphate and total phosphate with the Autoanalyzer. In: Technicon Symposia. Mediad Inc.: New York, pp. 385–389.
Berumen, M.L., Hoey, A.S., Bass, W.H., Bouwmeester, J., Catania, D., Cochran, J.E.M., Khalil, M.T., Miyake, S., Mughal, M.R., et al., 2013. The status of coral reef ecology research in the Red Sea. Coral Reefs. http://dx.doi.org/10.1007/s00338-013-1055-8.
Bongaerts, P., Hoeksema, B.W., Hay, K.B., Hoegh-Guldbery, O., 2012. Mushroom corals overcome live burial through pulsed inflation. Coral Reefs 31, 399.
Brown, O.B., Minnett, P.J., 1999. MODIS Infrared Sea Surface Temperature Algorithm Theoretical Basis Document, Ver 2.0. http://modis.gsfc.nasa.gov/data/atbd/atbd_mod25.pdf.
Brugère, C., Ridler, N., 2004. Global Aquaculture Outlook in the Next Decades: an Analysis of National Aquaculture Production Forecasts to 2030. FAO Fisheries Circular. No. 1001, Rome, FAO.
Bruno, J.F., Petes, L.E., Harvell, C.D., Hettinger, A., 2003. Nutrient enrichment can increase the severity of coral diseases. Ecol. Lett. 6, 1056–1061.
Bruno, J.F., Selig, E.R., Casey, K.S., Page, C.A., Willis, B.L., et al., 2007. Thermal stress and coral cover as drivers of coral disease outbreaks. PLoS Biol. 5, e124 http://dx.doi.org/10.1371/journal.pbio.0050124.
Central Department of Statistics, Kingdom of Saudi Arabia. http://www.citypopulation.de/SaudiArabia.html. (accessed 03.30.14.).
Cervino, J.M., Hayes, R., Polson, S., Polson, S.C., Goreau, T.J., Martinez, R.J., Smith, G.W., 2004a. Relationship of *Vibrio* species infection and elevated temperatures to yellow blotch/band disease in Caribbean corals. Appl. Environ. Microbiol. 70, 6855–6864.
Cervino, J.M., Hayes, R., Goreau, T.J., Smith, G.W., 2004b. Zooxanthellae regulation in yellow blotch/band and other coral disease contrasted with temperature-related bleaching: in situ destruction vs expulsion. Symbiosis 37, 63–85.
Cervino, J.M., Thompson, F.L., Gomez-Gil, B., Lorence, E.A., Goreau, T.J., Hayes, R.L., Winiarski-Cervino, K.B., Smith, G.W., Hughen, K., Bartels, E., 2008. The *Vibrio* core group induces yellow band disease in Caribbean and Indo-Pacific reef-building corals. J. App. Microbiol. 105, 1658–1671.
Cervino, J.M., Hauff, B., Haslun, J.A., Winiarski-Cervino, K., Cavazos, M., Lawther, P.,, Wier, A.M., Hughen, K., Strychar, K.B., 2012. Ulcerated yellow spot syndrome:

implications of aquaculture-related pathogens associated with soft coral *Sarcophyton ehrenbergi* tissue lesions. Dis. Aq. Org. 102, 137–148.
Chadwick, N., 1988. Competition and locomotion in free-living fungiid coral. J. Exp. Mar. Biol. Ecol. 123, 189–200.
Chadwick-Furman, N., Loya, Y., 1992. Migration, habitat use, and competition among mobile corals (Scleractinia: Fungiidae) in the Gulf of Eilat, Red Sea. Mar. Biol. 114, 617–623.
Cox, E.F., Ribes, M., Kinzie III, R.A., 2006. Temporal and spatial scaling of planktonic responses to nutrient inputs into a subtropical embayment. Mar. Ecol. Prog. Ser. 324, 19–35.
Dinsdale, E.A., Pantos, O., Smriga, S., Edwards, R.A., Angly, F., Wegley, L., Hatay, M., Hall, D., Brown, E., Haynes, M., Krause, L., Sala, E., Sandin, S.A., Thurber, R.V., Willis, B.L., Azam, F., Knowlton, N., Rohwer, F., 2008. Microbial ecology of four coral atolls in the Northern Line Islands. PLoS ONE 3, e1584. http://dx.doi.org/10.1371/journal.pone.001584.
Duedall, I.W., Bowman, M.J., O'Connors, H.B., 1975. Sewage sludge and ammonium concentrations in the New York Bight apex. Estuar. Coast. Mar. Sci. 3, 457–463.
Edmunds, P.J., Bruno, J.F., 1996. The importance of sampling scale in ecology: kilometer-wide variation in coral communities. Mar. Ecol. Prog. Ser. 143, 165–171.
Edmunds, P.J., 2002. Long-term dynamics of coral reefs in St. John, US Virgin Islands. Coral Reefs 21, 357–367.
Erftemeijer, P.L.A., Riegl, B., Hoeksema, B.W., Todd, P.A., 2012. Environmental impacts of dredging and other sediment disturbances on corals: a review. Mar. Pollut. Bull. 64, 1737–1765.
Fabricius, K.E., 2005. Effects of terrestrial runoff on the ecology of corals and coral reefs: review and synthesis. Mar. Pollut. Bull. 50, 125–146.
Furby, K.A., Bouwmeester, J., Berumen, M.L., 2013. Susceptibility of central Red Sea corals during a major bleaching event. Coral Reefs 32, 505–513.
Garcia-Fernandez, J.M., de Marsac, N.T., Diez, J., 2004. Streamlined regulation and gene loss as adaptive mechanisms in *Prochlorococcus* for optimized nitrogen utilization in oligotrophic environments. Microbiol. Mol. Rev. 68, 630–638.
Garren, M., Smriga, S., Azam, F., 2008. Gradients of coastal fish farm effluents and their effect on coral reef microbes. Environ. Microbiol. 10, 2299–2312.
Gilmour, J.P., 2004. Asexual budding in fungiid corals. Coral Reefs 23, 595.
Goffredo, S., Chadwick-Furman, N., 2000. Abundance and distribution of mushroom corals (Scleractinia: Fungiidae) on a coral reef at Eilat, Northern Red Sea. Bull. Mar. Sci. 66, 241–254.
Goreau, T.J., Cervino, J., Goreau, M., Hayes, R., Hayes, M., Richardson, L., Smith, G., DeMeyer, K., Nagelkerken, I., Garzon-Ferrera, J., Gil, D., Garrison, G., Williams, E.H., Bunkley-Williams, L., Quirolo, C., Patternson, K., Porter, J.W., Porter, K., 1998. Rapid spread of diseases in Caribbean coral reefs. Rev. Biol. Trop. 46, 157–171.
Green, E.P., Bruckner, A.W., 2000. The significance of coral disease epizootiology for coral real conservation. Biol. Conserv. 96, 347–361.
Harvell, C.D., Kim, K., Burkholder, J.M., Colwell, R.R., Epstein, P.R., et al., 1999. Emerging marine diseases – climate links and anthropogenic factors. Science 285, 1505.
Harvell, C.D., Mitchell, C.E., Ward, J.R., Altizer, S., Dobson, A.P., et al., 2002. Climate warming and disease risks for terrestrial and marine biota. Science 296, 2158–2162.
Harvell, C.D., Jordan-Dahlgren, E., Merkel, S., Rosenberg, E., Raymundo, L., Smith, G., Weil, E., Willis, B., 2007. Coral disease, environmental drivers, and the balance between coral and microbial associates. Oceanography 20, 172–195.
Hayes, M.L., Bonaventura, J., Mitchell, T.P., Prospero, J.M., Shinn, E.A., Van Dolah, F., Barber, R.T., 2001. How are climate and marine biological outbreaks functionally linked? Hydrobiologia 460, 213–220.
Hoegh-Guldberg, O., 1999. Climate change, coral bleaching and the future of the world's coral reefs. Mar. Freshw. Res. 50, 839–866.
Hoekesema, B.W., 1989. Taxonomy, phylogeny and biogeography of mushroom corals (Scleractinia: Fungiidae). Zool. Verh. 254, 3–295.
Hoeksema, B.W., 1991. Control of bleaching in mushroom coral populations (Scleractinia: Fungiidae) in the Java Sea: stress tolerance and interference by life history strategy. Mar. Ecol. Prog. Ser. 74, 225–237.
Hoeksema, B.W., Matthews, J.L., 2011. Contrasting bleaching patterns in mushroom coral assemblages at Koh Tao, Gulf of Thailand. Coral Reefs 30, 95.
Hoeksema, B.W., 2012a. Distribution patterns of mushroom corals (Scleractinia: fungiidae) across the Spermonde shelf, south Sulawesi. Raffles Bull. Zool. 60, 183–212.
Hoeksema, B.W., 2012b. Evolutionary trends in onshore-offshore distribution patterns of mushroom coral species (Scleractinia: Fungiidae). Contrib. Zool. 81, 199–221.
Holmes, R.M., Aminot, A., Kerouel, R., Hooker, B.A., Peterson, B.J., 1999. A simple and precise method for measuring ammonium in marine and freshwater ecosystems. Can. J. Fish. Aq. Sci., 1801–1808.
Kushmaro, A., Rosenberg, E., Fine, M., Ben Haim, Y., Loya, Y., 1998. Effects of temperature on bleaching of the coral *Oculina patagonica* by *Vibrio* AK-1. Mar. Ecol. Prog. Ser. 171, 131–137.
McClanahan, T., Polunin, N., Done, T., 2002. Ecological states and the resilience of coral reefs. Conserv. Ecol. 6, 18.
Moore, L.R., Goericke, R., Chisholm, S.W., 1995. Comparative physiology of Synechococcus and *Prochlorococcus*: influence of light and temperature on growth, pigments, fluorescence and absorptive properties. Mar. Ecol. Prog. Ser. 116, 259–275.
Partensky, F., Blanchot, J., Vaulot, D., 1999. Differential distribution and ecology of *Prochlorococcus* and *Synechococcus* in oceanic waters : a review. Bull. l'Institut océanogr. NS19, 457–475.
Pastorok, R., Bilyard, G., 1985. Effects of sewage pollution on coral-reef communities. Mar. Ecol. Prog. Ser. 21, 175–189.
Peters, E.C., 1984. A survey of cellular reactions to environmental stress and disease in Caribbean scleractinian corals. Helgol. Meeresunters 37, 113–137.
Porter, J.W., Dustan, P., Jaap, W.C., Patterson, K.L., Kosmynin, V., Meier, O.W., Patterson, M.E., Parsons, M., 2001. Patterns of spread of coral disease in the Florida Keys. Hydrobiologia 460, 1.
Quadfasel, D., 2001. Red sea circulation. In: Steele, J.H., Thorpe, S.A., Turekian, K.K. (Eds.), Encyclopedia of Ocean Sciences. Academic Press, Oxford, pp. 666–676.
Risk, M.J., Lapointe, B.E., Sherwood, O.A., Bedford, B.J., 2009. The use of [delta]15N in assessing sewage stress on coral reefs. Mar. Pollut. Bull. 58, 793–802.
Rosenberg, E., Ben-Haim, Y., 2002. Minireview: microbial diseases of corals and global warming. Environ. Microbiol. 4, 318–326.
Ruangpan, L., Kitao, T., 1991. *Vibrio* bacteria isolated from black tiger shrimp, *Penaeus monodon* Fabricius. J. Fish. Dis. 14, 383–388.
Sato, Y., Bourne, D.G., Willis, B.L., 2009. Dynamics of seasonal outbreaks of black band disease in an assemblage of *Montipora* species at Pelorus Island (Great Barrier Reef, Australia). Proc. Royal Soc. B 276, 2795–2803.
Schumacher, H., 1977. Ability of fungiid corals to overcome sedimentation. In: Proceedings of the 3rd International Coral Reef Symposium. Rosentiel School of Marine and Atmospheric Science, University of Miami.
Selig, E.R., Harvell, C.D., Bruno, J.F., Willis, B.L., Page, C.A., Casey, K.S., Sweatman, H., 2006. Analyzing the relationship between ocean temperature anomalies and coral disease outbreaks at broad spatial scales. In: Phinney, J.T., Strong, A., Skrving, W., West, J., Kleypas, J., Hough- Guldberg, O. (Eds.), Coral Reefs and Climate Change: Science and Management, Coastal and Estuarine Series, vol. 61. American Geophysical Union Press, Washington, D.C, pp. 111–128.
Sheppard, C.R.C., 1979. Interspecific aggression between reef corals with reference to their distribution. Mar. Ecol. Prog. Ser. 1, 237–247.
Smith, S.V., Kimmerer, W.J., Laws, E.A., Brock, R.E., Walsh, T.W., 1981. Kaneohe Bay sewage diversion experiment: perspectives on ecosystem responses to nutritional perturbation. Pac. Sci. 35, 279–395.
Szmant, A., 2002. Nutrient enrichment on coral reefs: is it a major cause of coral reef decline? Estuaries 25, 743–766.
Toren, A., Landau, L., Kushmaro, A., Loya, Y., Rosenberg, E., 1998. Effects of temperature on adhesion of *Vibrio* strain AK-1 to *Oculina patagonica* and on coral bleaching. Appl. Environ. Microbiol. 64, 1379–1384.
U.S. Environmental Protection Agency, 1983. Nitrogen, Ammonia. Method 350.1 (Colorimetric, Automated, Phenate). Methods for Chemical Analysis of Water and Wastes: Cincinnati, pp. 350-1.1–350-1.4.
Vincent, P., 2003. Jeddah's environmental problems. Geogr. Rev. 93, 394–412.
Veron, J.E.N., 2000. Corals of the World. In: Stafford-Smith, M. (Ed.). Australian Institute of Marine Science, Townsville.
Walker, D.I., Ormond, R.F.G., 1982. Coral death from sewage and phosphate pollution at Aqaba, Red Sea. Mar. Pollut. Bull. 13, 21–25.
Walton, C.C., Pichel, W.G., Sapper, J.F., 1998. The development and operational application of nonlinear algorithms for the measurement of sea surface temperatures with the NOAA polar-orbiting environmental satellites. J. Geophys. Res. 103, 27999–28012.
Weisse, T., 1989. The microbial loop in the Red Sea: dynamics of pelagic bacteria and heterotrophic nanoflagellates. Mar. Ecol. Prog. Ser. 55, 241–250.
Wilkinson, C.R., Souter, D. (Eds.), 2008. Status of Caribbean Coral Reefs After Bleaching and Hurricanes in 2005. Global Coral Reef Monitoring Network, pp. 1–148.
Willis, B.L., Page, C.A., Dinsdale, E.A., 2004. Coral disease on the Great Barrier Reef. In: Rosenberg, E., Loya, Y. (Eds.), Coral Health and Disease. Springer, New York, pp. 69–104.
Work, T.M., Aeby, G.S., 2006. Systematically describing gross lesions in corals. Dis. Aq. Org. 70, 155–160.